

Aim of the study The paper set out to provide the first large-scale, quantitative comparison of exploratory testing (ET) with traditional test-case-based testing (TCT). Specifically, the authors wanted to determine whether ET yields higher effectiveness (number and nature of defects found) and efficiency (defects found per fixed time box) than TCT in realistic, manual functional-testing settings, thereby addressing the scarcity of rigorous evidence about ET in both industry and academia.

Methodology Four controlled experiments were run—one with master's students and three with industrial testers—using the open-source jEdit text editor seeded with 25 faults. A total of 70 subjects (24 practitioners, 46 top-performing students) were randomly assigned to ET or TCT groups, with identical 90-minute time-boxed sessions preceded by identical 15-minute briefings. Both groups applied the same core design techniques (equivalence partitioning, boundary values, combinations); only the level of pre-documentation differed. Primary response variables were the number of true defects, detection difficulty (FTFI modes 0-3), technical type, severity, and false-report count. The experiment employed a single-factor, two-treatment design and identical materials (user guide, charter/templates) for both groups.

Results Across all sessions, ET uncovered 292 true defects versus 62 for TCT and achieved a significantly higher mean defect count (8.34 vs 1.83 per tester, $p \approx 1 \times 10^{-10}$). ET also revealed proportionally more difficult (modes 2-3), performance-, missing-function- and technical-class defects, and more severe/critical problems, while the two approaches produced comparable numbers of false reports. No significant interaction was found between tester type (student vs practitioner) and approach, indicating that ET's advantage held in both academic and industrial sub-samples.

Implications for research and practice For practitioners operating under tight schedules, ET delivers markedly higher fault yield without the overhead of full test-case documentation, suggesting that organizations can boost early-cycle defect detection by allocating focused ET sessions and reserving detailed scripts for regression or compliance contexts. Test managers should, however, retain TCT where repeatability and traceability are paramount. For researchers, the study highlights fertile directions: measuring feature coverage and cost trade-offs of ET, integrating ET with automated techniques, quantifying how tester experience shapes ET outcomes, and studying ET in larger, safety-critical domains. Replications across diverse systems and tester populations are encouraged to build a cumulative evidence base.